

TOWSON UNIVERSITY
COLLEGE OF GRADUATE STUDIES

AGENTIC ENVIRONMENT SYNTHESIS: EVALUATING THE EFFICACY OF
MODEL CONTEXT PROTOCOL (MCP) IN REDUCING HALLUCINATIONS OF
DECLARATIVE SYSTEM CONFIGURATION.

by

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Presented to the faculty

of Towson University

in partial fulfillment

of the requirements for the degree

Master's of Computer Science

Department of Computer and Information Sciences

Towson University,

Towson, Maryland

21252

December



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I'd like to thank Dr. Taylor and Dr. Kaza for their support throughout my graduate program and for the opportunities they have presented. I would also like to thank my parents for their belief in my ability to succeed by with their support.

Abstract

AGENTIC ENVIRONMENT SYNTHESIS: EVALUATING THE EFFICACY OF MODEL CONTEXT PROTOCOL (MCP) IN REDUCING HALLUCINATIONS OF DECLARATIVE SYSTEM CONFIGURATION.

Luis Dale Gascon

Inconsistent development environment can slow down software developers and their team. A new hire spends days configuring their systems. Engineers are tasked in resolving environment issues instead of pushing features.

In an educational environment, having students configure their machines for each course can act as a significant blocker to actual learning. An entire lecture period is often sacrificed to the manual and imperative installation of software and dependencies. Personal devices introduce unknown variables such as their operating systems, configuration, or hardware, resulting in students having to be troubleshoot problems specific to their devices that puts an instructor as a role of tech support for devices they might not have experience with.

I propose a solution that leverages agentic LLMs to synthesize a nix environment via nix flakes. We first provide an instructor's natural language description, then the agent queries an mcp-nixos server to discover valid package names and configuration options from real time data. The output is validated via nix evaluation commands, where the functional nature of nix provides a deterministic

success or failure signal. In the event of a failure, the LLM uses the error trace as context to correct itself until a valid, reproducible environment is produced.

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Chapter I.

Introduction

A. Software Configuration Management

B. Nix

1) Nix Flakes:

C. Docker

1) DevContainers:

D. Agent Harness

1) MCP Servers:

E. Table of Contents, List of Tables, and List of Figures

Chapter II.

Related Works

A. DevContainers to Standardize Student Development Environments

[1]

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Chapter III.

Figures

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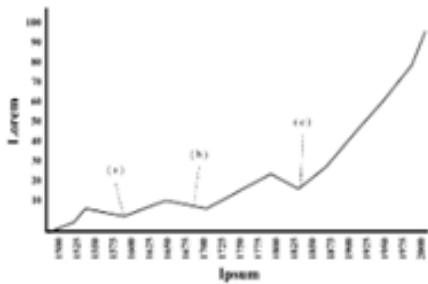


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Fig. 1. IEEE Style Figure Caption.

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Chapter IV.

Tables

Table I
IEEE Style Table Heading

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Qrstuv	9578	33

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* Notes that apply to a specific part of the table should be marked in the same manner as a footnote.

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Chapter V.

Testing Methodology

A. Agent flake synthesis testing

1) Pass@K:

B. Comparison between Nix and DevContainer hardware resource usage

Chapter VI.

Examples of Some Special Cases

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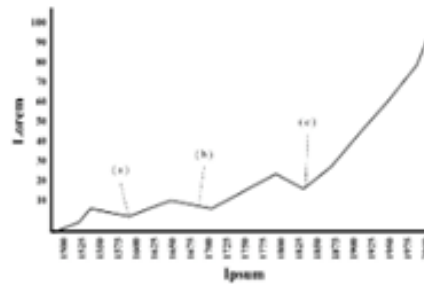


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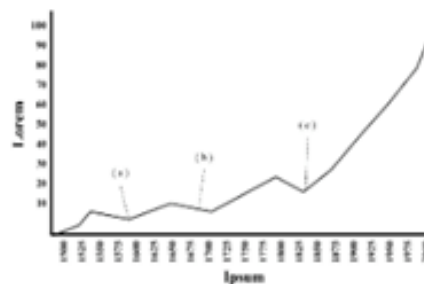


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You will find that the character s is isolated at the beginning of a new line. This should be avoid. The way to solve this is also included in the very beginning of Dissertation.tex. You can find the solution using `~` in the code editor in the next paragraph.

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Appendix A

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Appendix B

Appendix Title Here

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References

- [1] S. Valstar, W. G. Griswold, and L. Porter, “Using DevContainers to Standardize Student Development Environments: An Experience Report,” in Proceedings of the 2020 ACM Conference on Innovation and Technology in Computer Science Education. ACM, pp. 377–383. [Online]. Available: <https://dl.acm.org/doi/10.1145/3341525.3387424>

